VECTORS

A PHYSICAL QUANTITY THAT HAS BOTH MAGNITUDE AND DIRECTION E.G. FORCE

Introduction

Definitions of Vectors and Scalars

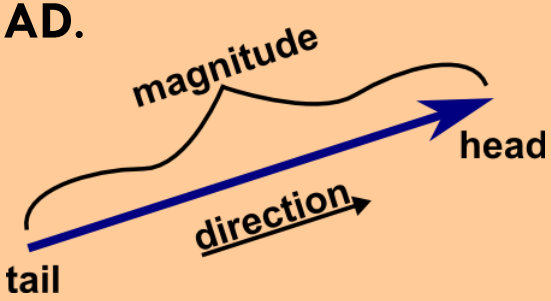
Physical quantities can be classified under two main headings -- Vectors and Scalars.

A **vector quantity** is any quantity that has **both magnitude (size) and direction.**
E.g., velocity, acceleration, force, momentum.

A **scalar quantity** is any quantity that has **magnitude only**, while direction is not taken into account.
E.g., speed, pressure, temperature, energy.

REPRESENTATION OF VECTORS

1. **A VECTOR CAN BE REPRESENTED BY A STRAIGHT LINE DRAWN TO SCALE WITH AN ARROWHEAD.**



- THE LENGTH REPRESENTS THE SIZE OF THE VECTOR
- THE ARROWHEAD REPRESENTS THE DIRECTION OF THE VECTOR
- THE STARTING POINT IS CALLED THE TAIL OF THE VECTOR AND THE END POINT IS CALLED THE HEAD

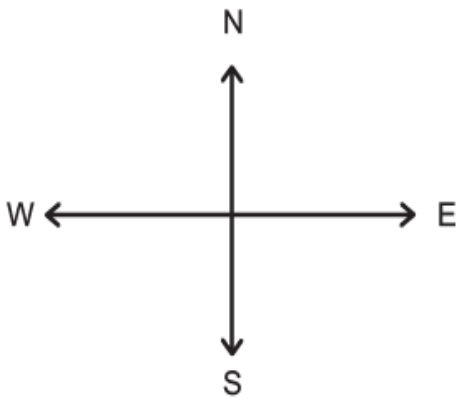
2. **VECTORS CAN BE REPRESENTED USING VECTOR NOTATION**

- THE SYMBOL OF THE VECTOR IS WRITTEN WITH AN ARROW ON TOP. THIS MEANS THAT THE VECTORS HAVE BOTH MAGNITUDE AND DIRECTION.

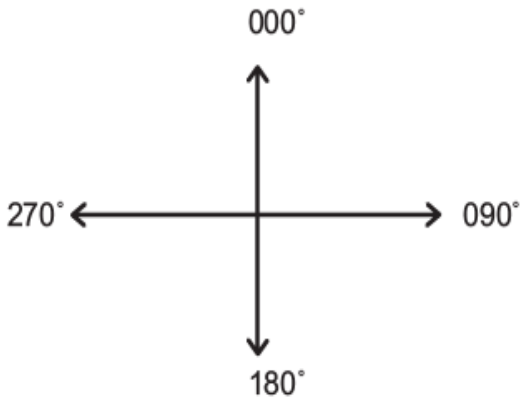
$\vec{F}$ → Refers to the force vector, both magnitude and direction

DIRECTION OF VECTORS

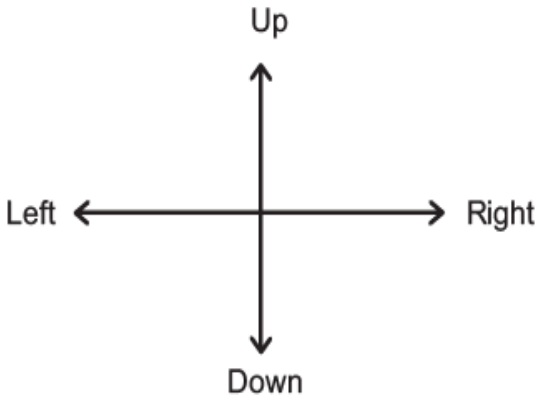
1. Cardinal points



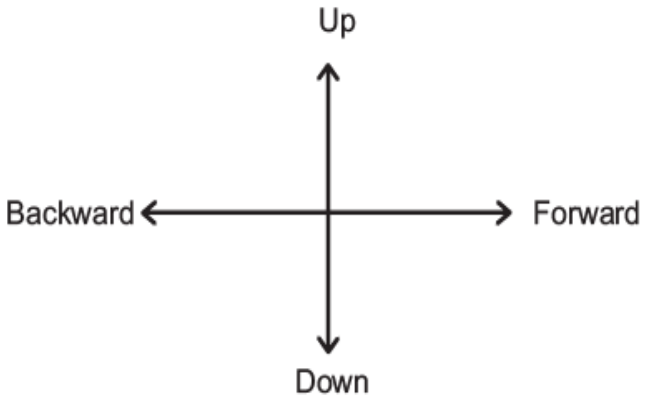
2. Bearing



3. Left & right, up & down



4. Forward, backward, up and down



ADDITION OF VECTORS

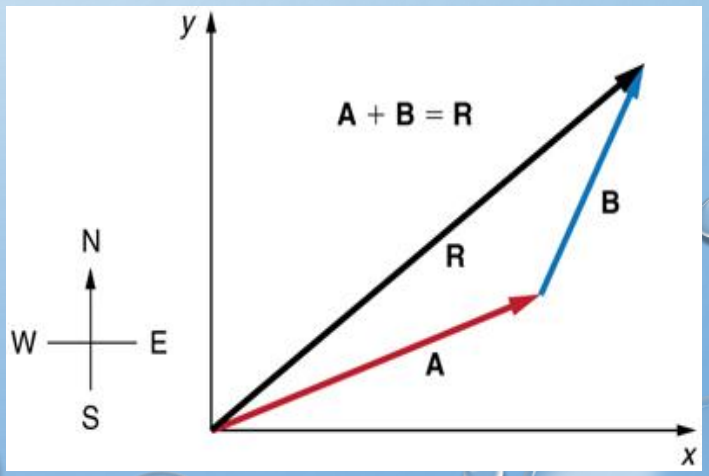
THE RESULTANT VECTOR – A SINGLE VECTOR WHICH HAS THE SAME EFFECT AS THE ORIGINAL VECTORS ACTING TOGETHER.

1. GRAPHICAL METHODS

   I HEAD TO TAIL METHOD (LEARNER BOOK PG 6)

   - CHOOSE A SCALE AND REFERENCE POINT.

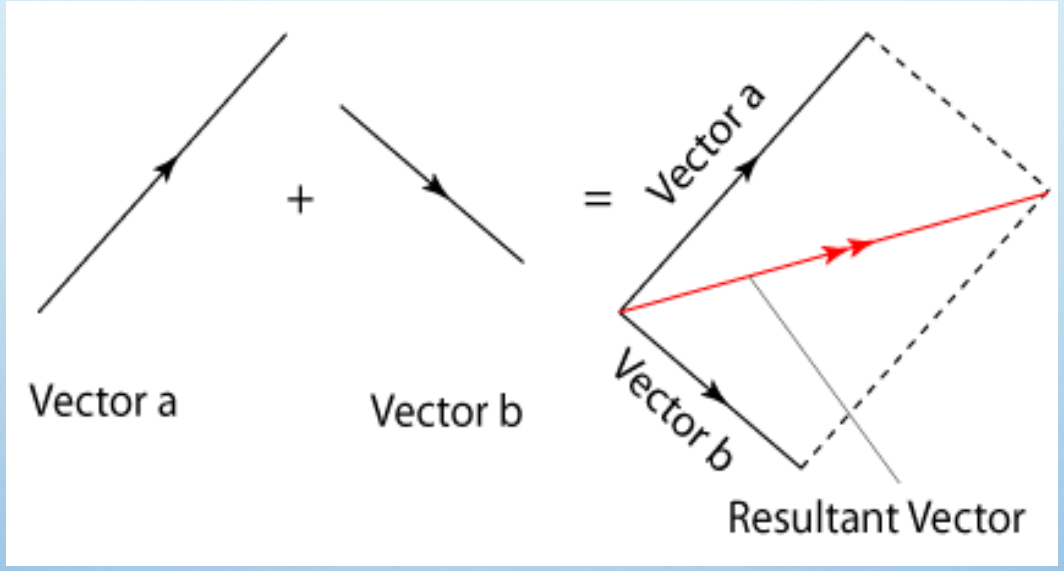
   - DRAW THE VECTORS HEAD TO TAIL AND DETERMINE THE RESULTANT VECTOR



II PARALLELOGRAM METHOD (LEARNER BOOK PG 6)

- A VECTOR REMAINS THE SAME AS LONG AS IT HAS THE SAME MAGNITUDE AND DIRECTION.

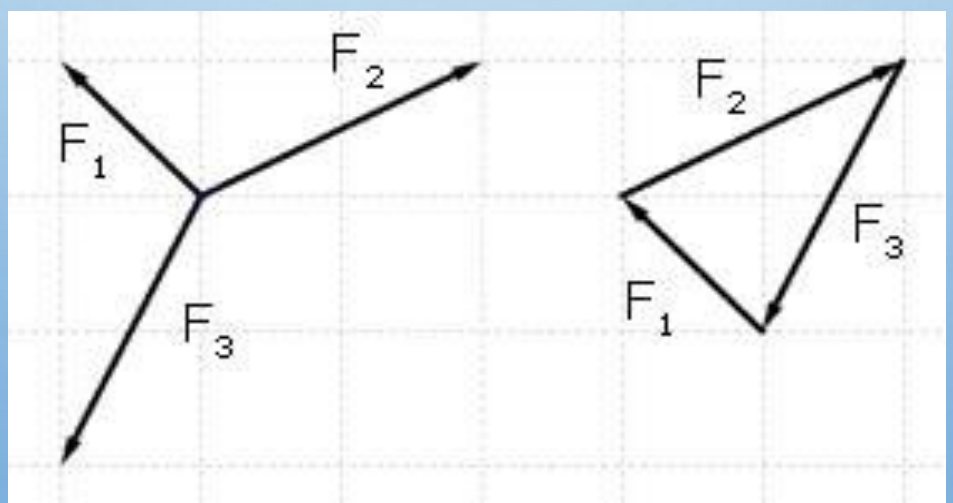
- VECTORS CAN BE REARRANGED TO FORM A PARALLELOGRAM



2. ALGEBRAIC METHODS – USES PYTHAGORAS FOR RIGHT-ANGLED TRIANGLES AND TRIG RATIOS.

VECTORS IN EQUILIBRIUM

- WHEN THE RESULTANT OF TWO OR MORE VECTORS EXERTED AT A POINT ARE EQUAL TO ZERO, THE VECTORS ARE IN EQUILIBRIUM.
- THE EQUILIBRANT IS THE SINGLE VECTOR THAT KEEPS THE OTHER FORCES IN BALANCE(EQUILIBRIUM). IT HAS THE SAME MAGNITUDE AS THE RESULTANT BUT ACTS IN THE OPPOSITE DIRECTION.
- WHEN VECTORS ACTING AT A POINT ARE IN EQUILIBRIUM, THEIR MAGNITUDE AND DIRECTION CAN BE REPRESENTED SEQUENTIALLY BY THE SIDES OF A TRIANGLE.



EXERCISE

FIND THE RESULTANT FORCE BY EITHER DRAWING A SCALE DIAGRAM OR USING ALGEBRAIC METHODS.

1. 25 N RIGHT; 40 N RIGHT
2. 36 N UP; 23 N UP; 4 N UP
3. 45 N UP; 65 N DOWN
4. 240 N LEFT; 20 N RIGHT
5. WEIGHT 68 N; UPWARD FORCE 68 N; FRICTIONAL FORCE 5 N TO THE RIGHT
6. DOWNWARD FORCE 25 N; 40 N HORIZONTAL FORCE TO THE LEFT

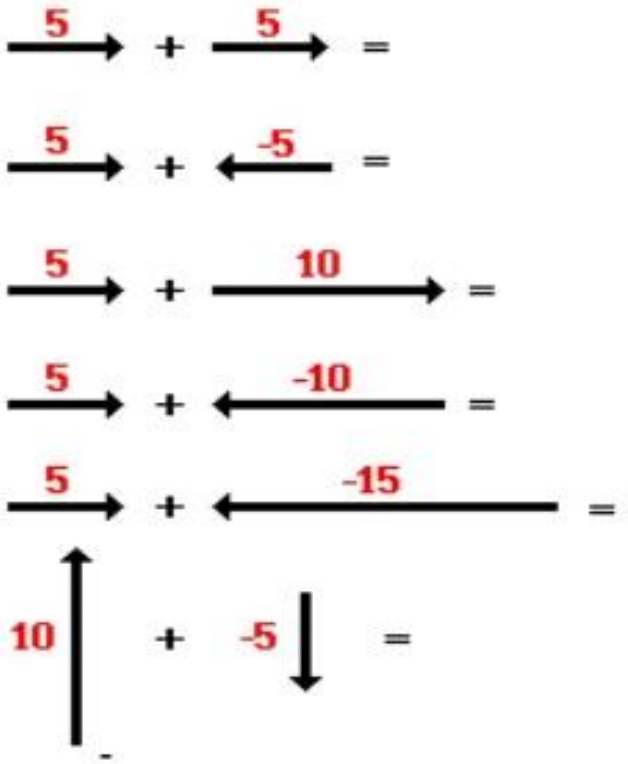
RESULTANT VECTOR

- THE RESULTANT IS THE SINGLE VECTOR THAT HAS THE SAME EFFECT AS TWO OR MORE VECTORS TOGETHER.
- THE EQUILIBRANT IS THE VECTOR THAT HAS THE SAME MAGNITUDE BUT ACTS IN THE OPPOSITE DIRECTION TO THE RESULTANT.

Two vectors can be added together to determine the result (or resultant).

- The two methods used throughout the entire unit are:

1. the Pythagorean theorem and trigonometric methods
2. the head-to-tail method using a scaled vector diagram



PROPERTIES OF VECTORS

1. **EQUALITY OF VECTORS:-** VECTORS ARE EQUAL IF THEY HAVE THE SAME SIZE AND ARE IN THE SAME DIRECTION.
2. **NEGATIVE VECTORS:-** A NEGATIVE VECTOR IS A VECTOR WITH THE SAME SIZE BUT THAT ACTS IN THE OPPOSITE DIRECTION TO THE GIVEN POSITIVE DIRECTION.
3. **ADDITION OF VECTORS:-** THE SUM OF TWO VECTORS PRODUCES A NEW VECTOR WHERE THE SIZE AND DIRECTION DEPENDS ON THE TWO ORIGINAL VECTORS.
4. **SUBTRACTION OF VECTORS:-** THIS IS THE SAME AS ADDING A POSITIVE VECTOR TO A NEGATIVE VECTOR.
5. **MULTIPLICATION OF A VECTOR:-** A VECTOR CAN BE MULTIPLIED BY A SCALAR QUANTITY. IF THE SCALAR IS POSITIVE, THE DIRECTION REMAINS THE SAME. IF THE SCALAR IS NEGATIVE, THE DIRECTION IS OPPOSITE THE ORIGINAL VECTOR.